JulleyOnline Collaborative Kanban — REST API Documentation

Base URL: `http://localhost:5000/api`
Authentication: Bearer JWT in `Authorization` header (`Authorization: Bearer <token>`)
Tenant Scoping: Active branch UUID supplied in `x-tenant-id` header

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1. Authentication APIs

Register User

- **Method:** `POST`
- **Endpoint:** `/api/auth/register`
- **Authentication:** Public
- **Request Body:**

```
{
  "name": "Arjun Sharma",
  "email": "arjun@abc.com",
  "password": "password123"
}
```

- **Responses:**
- `201 Created` : `{ "user": { "id": "uuid", "name": "...", "email": "..." }, "token": "jwt_token" }`
- `400 Bad Request` : `{ "error": "Email is already registered." }`

Login User

- **Method:** `POST`
- **Endpoint:** `/api/auth/login`
- **Authentication:** Public
- **Request Body:**

```
{
  "email": "rahul@abc.com",
  "password": "password123"
}
```

- **Responses:**

- `200 OK`: `{ "user": { "id": "uuid", "name": "Rahul", "email": "rahul@abc.com", "memberships": [...] }, "token": "jwt_token" }`
- `401 Unauthorized`: `{ "error": "Invalid email or password." }`

Get Current User Profile

- **Method:** `GET`
- **Endpoint:** `/api/auth/me`
- **Authentication:** `Bearer <token>`
- **Responses:**
- `200 OK`: `{ "user": { "id": "uuid", "name": "Rahul", "email": "rahul@abc.com", "memberships": [...] } }`

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2. Organization APIs

List Authorized Organizations

- **Method:** `GET`
- **Endpoint:** `/api/organizations`
- **Authentication:** `Bearer <token>`
- **Responses:**
- `200 OK`: `[ { "id": "uuid", "name": "ABC Technologies", "tenants": [...] } ]`

Get Organization by ID

- **Method:** `GET`
- **Endpoint:** `/api/organizations/:id`
- **Authentication:** `Bearer <token>`
- **Responses:**
- `200 OK`: `{ "id": "uuid", "name": "ABC Technologies", "tenants": [...], "memberships": [...] }`
- `404 Not Found`: `{ "error": "Organization not found or access denied." }`

Create Organization

- **Method:** `POST`
- **Endpoint:** `/api/organizations`
- **Authentication:** `Bearer <token>`
- **Request Body:** `{ "name": "New Enterprise Corp" }`
- **Responses:**

- `201 Created`: `{ "id": "uuid", "name": "New Enterprise Corp" }`

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3. Tenant (Branch) APIs

List Authorized Tenants

- **Method:** `GET`
- **Endpoint:** `/api/tenants?organization_id=:orgId`
- **Authentication:** `Bearer <token>`
- **Tenant-Scoping:** Only returns branches the user belongs to (or all branches if user is Org Admin).
- **Responses:**
- `200 OK`: `[ { "id": "uuid", "name": "Hyderabad", "organization_id": "uuid" }, ... ]`

Create Tenant Branch

- **Method:** `POST`
- **Endpoint:** `/api/tenants`
- **Authentication:** `Bearer <token>` (Required Role: `Org Admin`)
- **Request Body:** `{ "name": "Chennai", "organization_id": "uuid" }`
- **Responses:**
- `201 Created`: `{ "id": "uuid", "name": "Chennai" }`
- `403 Forbidden`: `{ "error": "Insufficient permissions. Required role: Org Admin" }`

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4. Board APIs

List Boards for Tenant

- **Method:** `GET`
- **Endpoint:** `/api/boards`
- **Headers:** `Authorization: Bearer <token>`, `x-tenant-id: <tenant_id>`
- **Tenant-Scoping:** Strictly filters boards where `tenant_id === header.x-tenant-id`.
- **Responses:**
- `200 OK`: `[ { "id": "uuid", "name": "Website Development", "columns": [...], "creator": {...} } ]`

Create Board

- **Method:** `POST`
- **Endpoint:** `/api/boards`

- **Headers:** `Authorization: Bearer <token>`, `x-tenant-id: <tenant_id>`
- **Required Role:** `Org Admin` or `Tenant Admin`
- **Request Body:**

```
{
  "name": "Mobile Application Development",
  "description": "iOS and Android apps",
  "organization_id": "uuid",
  "tenant_id": "uuid"
}
```

- **Responses:**
- `201 Created`: `{ "id": "uuid", "name": "Mobile Application Development", "columns": [...] }` (Automatically creates default 4 columns: To Do, In Progress, Testing, Done).

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5. Column APIs

Get Columns for Board

- **Method:** `GET`
- **Endpoint:** `/api/boards/:boardId/columns`
- **Headers:** `Authorization: Bearer <token>`, `x-tenant-id: <tenant_id>`
- **Responses:**
- `200 OK`: `[ { "id": "uuid", "name": "To Do", "position": 1, "tasks": [...] }, ... ]`

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6. Task APIs (With Optimistic Concurrency & Transactions)

Create Task

- **Method:** `POST`
- **Endpoint:** `/api/tasks`
- **Headers:** `Authorization: Bearer <token>`, `x-tenant-id: <tenant_id>`
- **Request Body:**

```
{
  "title": "Fix Authentication Bug",
  "description": "Handle token expiration gracefully",
  "board_id": "uuid",
  "column_id": "uuid",
  "tenant_id": "uuid",
  "assigned_to": "uuid"
}
```

- **Responses:**
- `201 Created` : Returns task with `version: 1` , calculated `position` , and assignee info.

Update Task (Optimistic Concurrency Check)

- **Method:** `PUT`
- **Endpoint:** `/api/tasks/:id`
- **Headers:** `Authorization: Bearer <token>` , `x-tenant-id: <tenant_id>`
- **Request Body:**

```
{
  "title": "Fix Authentication Bug (Resolved)",
  "description": "Updated fix details",
  "expected_version": 1
}
```

- **Responses:**
- `200 OK` : `{ "id": "uuid", "title": "...", "version": 2 }`
- `409 Conflict` :

```
{
  "error": "Conflict: Task was modified by another user. Current server version is 2, you
}
```

Move Task (Drag-and-Drop Database Transaction)

- **Method:** `PATCH`
- **Endpoint:** `/api/tasks/:id/move`
- **Headers:** `Authorization: Bearer <token>` , `x-tenant-id: <tenant_id>`
- **Request Body:**

```
{
  "destination_column_id": "uuid",
  "new_position": 2,
  "expected_version": 1
}
```

- **Behavior:** Inside a single database transaction, validates destination column, reorders surrounding tasks, increments task `version = version + 1`, logs activity, and broadcasts `task:moved` event to `tenant:<tenant_id>` room.
- **Responses:**
- `200 OK`: `{ "id": "uuid", "column_id": "...", "position": 2, "version": 2 }`
- `409 Conflict`: Stale drag-and-drop detected; returns HTTP 409 so UI reconciles with authoritative server state.

Delete Task

- **Method:** `DELETE`
- **Endpoint:** `/api/tasks/:id`
- **Headers:** `Authorization: Bearer <token>`, `x-tenant-id: <tenant_id>`
- **Responses:**
- `200 OK`: `{ "success": true, "message": "Task deleted successfully." }`